

Linear Correlation Coefficient

The correlation coefficient computed from the sample data measures the strength and direction of a linear relationship between two variables. The symbol for the sample correlation coefficient is r . The symbol for the population correlation coefficient is ρ (Greek letter rho).

Correlation Coefficient The measure ρ of linear association between two variables X and Y is estimated by the **sample correlation coefficient** r , where

$$r = b_1 \sqrt{\frac{S_{xx}}{S_{yy}}} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

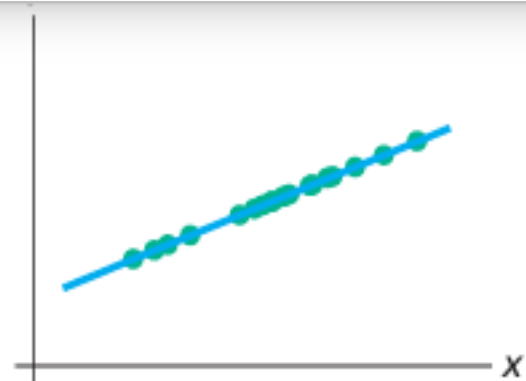
$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2} \sqrt{\sum(Y - \bar{Y})^2}}$$

or

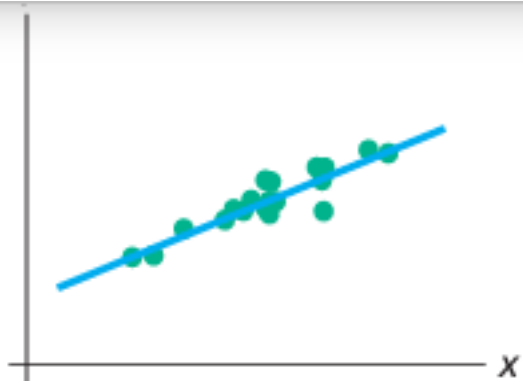
$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

The correlation coefficient lies between -1 to $+1$ i.e. $-1 \leq r \leq +1$

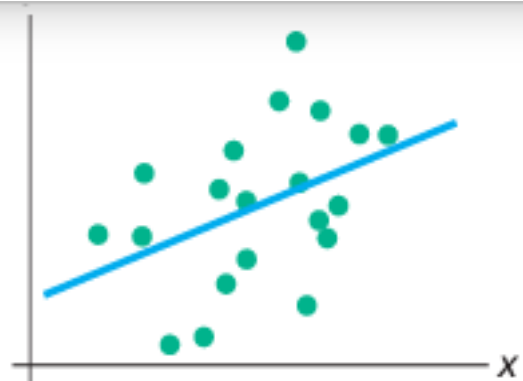
- If $\rho = 0$, the variables are **linearly uncorrelated**, meaning that there is no linear relationship between the variables.
- If $\rho \neq 0$, the variables are **linearly correlated**. Linearly correlated variables are either positively linearly correlated or negatively linearly correlated.
- If $\rho > 0$, the variables are **positively linearly correlated**, meaning that y tends to increase linearly as x increases (and vice versa), with the tendency being greater the closer ρ is to 1.
- If $\rho < 0$, the variables are **negatively linearly correlated**, meaning that y tends to decrease linearly as x increases (and vice versa), with the tendency being greater the closer ρ is to -1 .



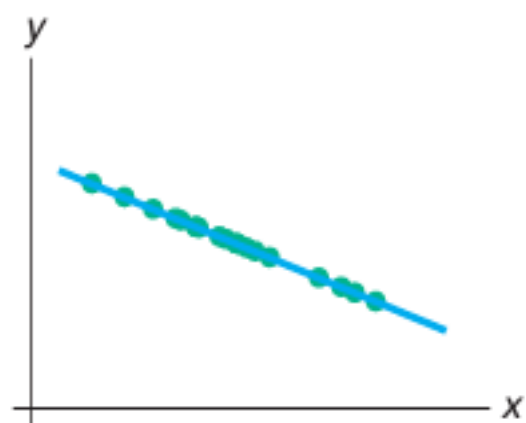
(a) Perfect positive
linear correlation
 $r = 1$



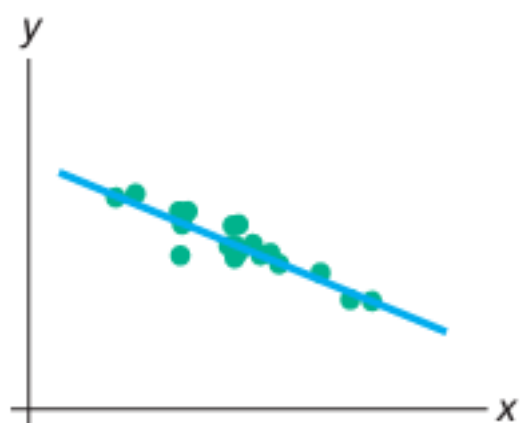
(b) Strong positive
linear correlation
 $r = 0.9$



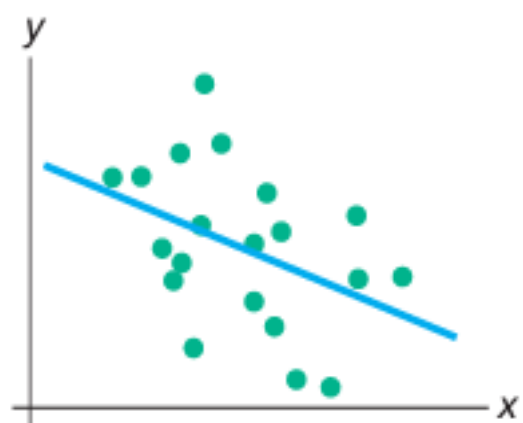
(c) Weak positive
linear correlation
 $r = 0.4$



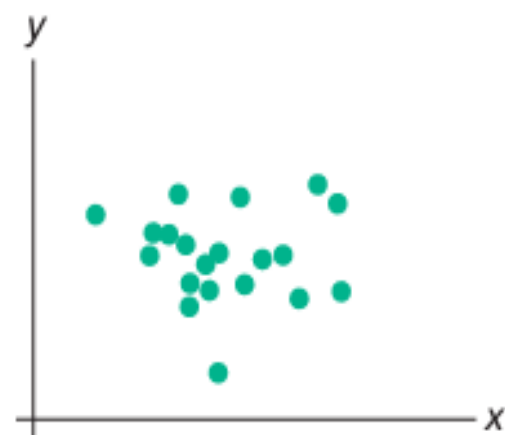
(d) Perfect negative
linear correlation
 $r = -1$



(e) Strong negative
linear correlation
 $r = -0.9$



(f) Weak negative
linear correlation
 $r = -0.4$



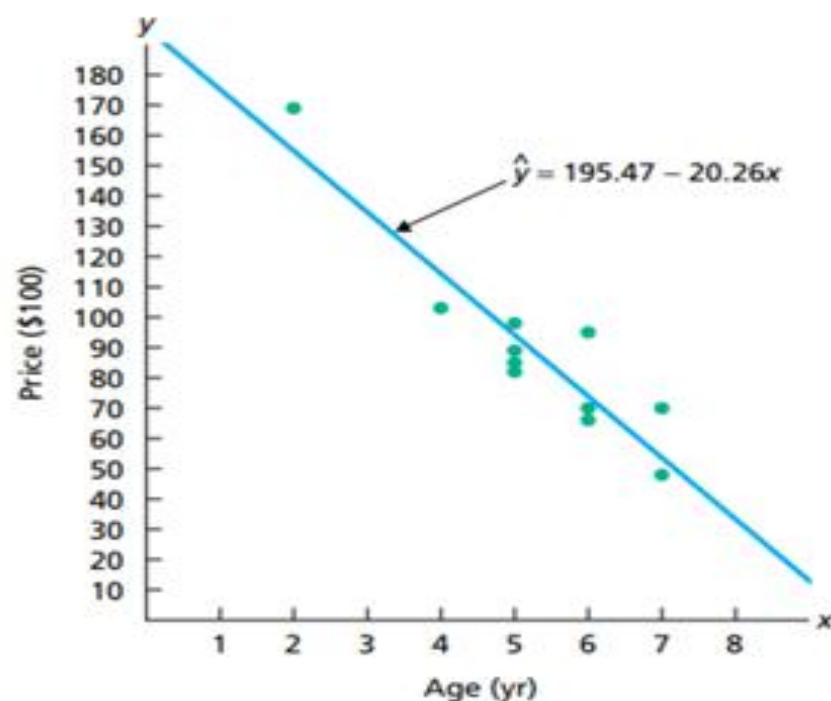
(g) No linear correlation
(linearly uncorrelated)
 $r = 0$

Example:

Age and Price of Orions The age and price data for a sample of 11 Orions are repeated in the first two columns of Table 14.10 on the next page.

- Compute the linear correlation coefficient, r , of the data.
- Interpret the value of r obtained in part (a) in terms of the linear relationship between the variables age and price of Orions.
- Discuss the graphical implications of the value of r .

Age (yr) x	Price (\$100) y	xy	x^2	y^2
5	85	425	25	7,225
4	103	412	16	10,609
6	70	420	36	4,900
5	82	410	25	6,724
5	89	445	25	7,921
5	98	490	25	9,604
6	66	396	36	4,356
6	95	570	36	9,025
2	169	338	4	28,561
7	70	490	49	4,900
7	48	336	49	2,304
58	975	4732	326	96,129



$$\begin{aligned}
 r &= \frac{\Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)/n}{\sqrt{[\Sigma x_i^2 - (\Sigma x_i)^2/n][\Sigma y_i^2 - (\Sigma y_i)^2/n]}} \\
 &= \frac{4732 - (58)(975)/11}{\sqrt{[326 - (58)^2/11][96,129 - (975)^2/11]}} = -0.924.
 \end{aligned}$$

- b. Interpretation** The linear correlation coefficient, $r = -0.924$, suggests a strong negative linear correlation between age and price of Orions. In particular, it indicates that as age increases, there is a strong tendency for price to decrease, which is not surprising. It also implies that the regression equation, $\hat{y} = 195.47 - 20.26x$, is extremely useful for making predictions.
- c.** Because the correlation coefficient, $r = -0.924$, is quite close to -1 , the data points should be clustered closely about the regression line. Figure 14.16 on page 663 shows that to be the case. 

Test statistic for correlation co-efficient

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}},$$

which, as before, is a value of the statistic T having a t -distribution with $n - 2$ degrees of freedom.

Correlation t-Test

Purpose To perform a hypothesis test for a population linear correlation coefficient, ρ

Assumptions The four assumptions for regression inferences

Step 1 The null hypothesis is $H_0: \rho = 0$, and the alternative hypothesis is

$$\begin{array}{ccccc} H_a: \rho \neq 0 & & H_a: \rho < 0 & & H_a: \rho > 0. \\ \text{(Two tailed)} & \text{or} & \text{(Left tailed)} & \text{or} & \text{(Right tailed)} \end{array}$$

Step 2 Decide on the significance level, α .

Step 3 Compute the value of the test statistic

$$t = \frac{r}{\sqrt{\frac{1 - r^2}{n - 2}}}$$

and denote that value t_0 .

CRITICAL-VALUE APPROACH

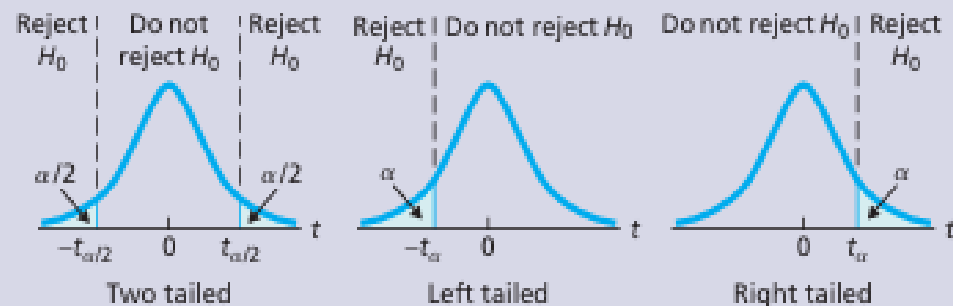
OR

P-VALUE APPROACH

Step 4 The critical value(s) are

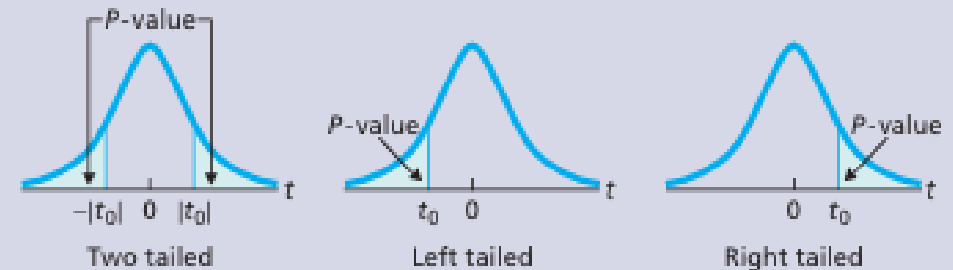
$\pm t_{\alpha/2}$ (Two tailed) or $-t_{\alpha}$ (Left tailed) or t_{α} (Right tailed)

with $df = n - 2$. Use Table IV to find the critical value(s).



Step 5 If the value of the test statistic falls in the rejection region, reject H_0 ; otherwise, do not reject H_0 .

Step 4 The t -statistic has $df = n - 2$. Use Table IV to estimate the P -value, or obtain it exactly by using technology.



Step 5 If $P \leq \alpha$, reject H_0 ; otherwise, do not reject H_0 .

Step 6 Interpret the results of the hypothesis test.

Example:

Age and Price of Orions The data on age and price for a sample of 11 Orions are repeated in Table 15.5. At the 5% significance level, do the data provide sufficient evidence to conclude that age and price of Orions are negatively linearly correlated?

TABLE 15.5

Age and price data
for a sample of 11 Orions

Age (yr) x	Price (\$100) y
5	85
4	103
6	70
5	82
5	89
5	98
6	66
6	95
2	169
7	70
7	48

Age (yr) x	Price (\$100) y	xy	x^2	y^2
5	85	425	25	7,225
4	103	412	16	10,609
6	70	420	36	4,900
5	82	410	25	6,724
5	89	445	25	7,921
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6	66	396	36	4,356
6	95	570	36	9,025
2	169	338	4	28,561
7	70	490	49	4,900
7	48	336	49	2,304
58	975	4732	326	96,129

$H_0: \rho = 0$ (age and price are linearly uncorrelated)

$H_a: \rho < 0$ (age and price are negatively linearly correlated).

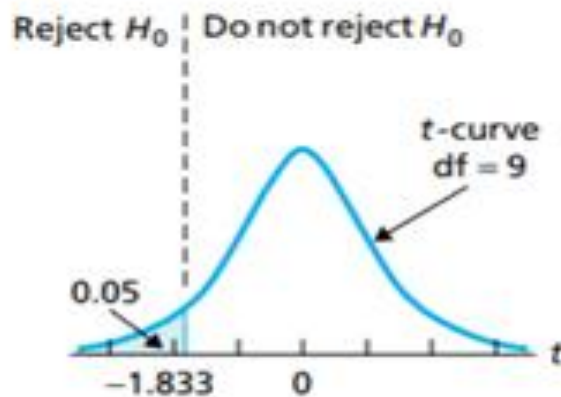
$\alpha = 0.05$.

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

$$t = \frac{-0.924}{\sqrt{\frac{1-(-0.924)^2}{11-2}}} = -7.249.$$

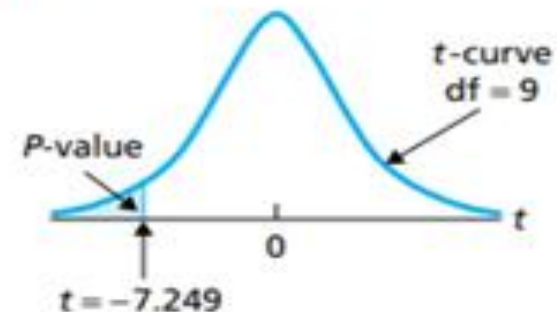
For $n = 11$, $df = 9$. Also, $\alpha = 0.05$. From Table IV, for $df = 9$, $t_{0.05} = 1.833$. Consequently, the critical value is $-t_{0.05} = -1.833$, as shown in Fig. 15.12A.

FIGURE 15.12A



From Step 3, the value of the test statistic is $t = -7.249$. Because the test is left tailed, the P -value is the probability of observing a value of t of -7.249 or less if the null hypothesis is true. That probability equals the shaded area shown in Fig. 15.12B.

FIGURE 15.12B



The value of the test statistic, found in Step 3, is $t = -7.249$. Figure 15.12A shows that this value falls in the rejection region, so we reject H_0 . The test results are statistically significant at the 5% level.

Step 5 If $P \leq \alpha$, reject H_0 ; otherwise, do not reject H_0 .

From Step 4, $P < 0.005$. Because the P -value is less than the specified significance level of 0.05, we reject H_0 . The test results are statistically significant at the 5% level and (see Table 9.8 on page 408) provide very strong evidence against the null hypothesis.

A Measure of Quality of Fit: Coefficient of Determination

This quantity is a measure of the proportion of variability explained by the fitted model.

$$R^2 = (r)^2$$

Note: The coefficient of determination, r^2 , always lies between 0 and 1. A value of r^2 near 0 suggests that the regression equation is not very useful for making predictions, whereas a value of r^2 near 1 suggests that the regression equation is quite useful for making predictions.

$$SST = SSR + SSE$$

CONNECTION?

Total variability = Explained variability + Unexplained variability

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n e_i^2$$

